

- I. PROGRAM TITLE: Preserving Life Through First Aid - A STEM Approach for Non-Health Providers
- II. PROPONENT: JENNIFER P. BENLIRO, DPA
CO- PROPONENT: Rizza Almalbis, Daisy Mae Bongtiwon, Jay Ann Gregorio, Kathleen Patrona, Joesan Borres, Jan Rosarth Banilla
- III. RATIONALE/BACKGROUND:

Emergencies and accidents can occur unexpectedly, regardless of time or location. Providing immediate and appropriate first aid is crucial for improving survival rates and minimizing injury severity. By integrating STEM principles into first aid training, our goal is to enhance the learning experience, making it more engaging, effective, and accessible to a wider audience.

The initiative, "Preserving Life Through First Aid," is an innovative project designed to equip non-health providers with essential first aid skills using a Science, Technology, Engineering, and Mathematics (STEM) approach. The project seeks to empower individuals with the knowledge and tools to respond effectively to emergencies, ultimately increasing the likelihood of preserving life in critical situations.

IV. OBJECTIVES

Emergency situations require immediate and effective first aid responses, yet non-health providers often lack the necessary skills. This project aims to address this gap through a STEM-based first aid training program. Specifically, it seeks to answer the following questions:

1. How can STEM principles be integrated into a comprehensive first aid training program?
2. What educational materials and technological aids are most effective in facilitating learning and retention?
3. How do hands-on training sessions impact the response time and effectiveness of non-health providers in emergency situations?
4. What is the level of awareness and understanding of basic first aid principles among non-health providers before and after the training program?

V. ACTIVITIES/METHODOLOGY

The project not only teaches people about first aid, but it also hopes to encourage them to modify their behavior in health-related circumstances. In conclusion, the project's first aid training curriculum is built on a teaching method that incorporates active experiences, STEM principles, and technology. This teaching technique is supported by the emphasis on health education and behavior modification, which emphasizes participants' active roles in knowledge construction through meaningful interactions with their training environment.

Variables

1. Independent Variables: STEM-based first aid training program, educational materials, technological aids, and hands-on training sessions.

2. Dependent Variables: Awareness and understanding of first aid principles, response time, effectiveness in emergency situations.

- **Design Development:**

Work with experts in both first aid and STEM fields to develop a design that integrates scientific principles into practical first aid techniques.

- **Educational Materials:**

Create multimedia content, including videos, interactive simulations, and instructional guides, to facilitate learning and retention.

- **Training Sessions:**

Organize hands-on training sessions using technological aids such as virtual reality and simulations to provide realistic scenarios for practice.

VI. WORKPLAN/GANTT CHART

The project is expected to be completed within 18 months, with specific milestones for program development, content creation, pilot training sessions, and outreach activities.

We aim to collaborate with local schools, community centers, and relevant organizations to ensure widespread dissemination of the first aid training program.

Partner Agencies:

Philippine Red Cross - They will provide the specific first-aid training materials

2. EARIST-MANILA (c/o/ Daisy Mae Bongtiwon) – She will provide a seminar about STEM and its application to Health and Clinical sciences.

3. CAPIZ STATE UNIVERSITY & EARIST – Dr. Jennifer Benliro and Daisy Mae Bongtiwon, et. al. will design a training materials such as modules and manuals on first-aid training incorporating STEM skills.

4. Local Health Unit – They will provide the participants for this activity.

If possible, regional level so that the training will be used by the entire health system.

Research Objectives	2024				2025		
	Jan	Feb - May	June-Aug	Sept-Dec.	Jan -- Mar	Apr - June	Jul – Dec.
Proposal Presentation							
Program Design Development							
Educational Materials Creation							
Training Sessions							
Evaluation/ Impact Study							
Presentation to Symposia							

VII. OUTPUTS AND IMPACTS

1. STEM Literacy Enhancement:

Skills Acquisition: Develop a foundational understanding of STEM principles, terminology, and applications relevant to their roles.

2. Critical Thinking Development:

- Skills Acquisition: Enhance critical thinking skills through scenario-based training and exposure to diverse emergency situations.

3. Creativity in First Aid Response:

- Skills Acquisition: Promote creativity through training exercises that challenge conventional problem-solving approaches.

4. Effective Communication Skills:

- Skills Acquisition: Develop effective verbal and non-verbal communication skills, including empathy and active listening.

5. Collaborative Teamwork:

- Skills Acquisition: Foster teamwork through simulations and exercises that require coordinated efforts in emergency response scenarios.

6. Adaptability and Technological Proficiency:

- Skills Acquisition: Provide training on the use of modern technologies and tools relevant to first aid, such as automated external defibrillators (AEDs) or telemedicine applications.

7. Cultural Competence and Sensitivity:

- Skills Acquisition: Offer cultural competence training to enhance sensitivity and responsiveness to the needs of individuals from various cultural and social contexts.

8. Continuous Learning and Skill Refinement:

- Skills Acquisition: Encourage participation in regular training programs, workshops, and educational opportunities to stay updated on the latest advancements in first aid and STEM-related knowledge.

By aligning their objectives and skill acquisition efforts with these key areas, non-health first aid providers can enhance their effectiveness in responding to emergencies, contribute to the overall healthcare system, and ensure the well-being of those they serve

VIII. BUDGET

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		150,000
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		150,000
- Meal allowance during data gathering		50,000
- Professional fees (laboratory analysis and other related fees)		80,000
- Miscellaneous (for incidental expenses)		50,000
- Training Sessions		150,000
- Printing Materials		75,000

C. CAPITAL OUTLAY Printing Equipment		150,000
Total		855,000.00

References:

1. Roshchin, A. L. (2010). The importance of experiential learning in first aid: Collaborations between nursing schools and healthcare agencies.
2. Bail, C. (2021). Enhancing first aid knowledge in engineering students: A theoretical-practical approach.
3. Evens, M. (2010). Widespread first aid training as a social responsibility.
4. Tunnicliffe, S. D. (2007). Integrating first aid into science education: A proposal for curriculum enhancement.
5. Recio, J. M. (1985). Enhancing first aid knowledge and training among non-health providers in the Philippines.
6. Ho, A. (2018). Addressing health challenges among non-health providers in the Philippines: The imperative for improved first aid knowledge and training.
7. Garces, L. (2019). Innovative approaches to first aid instruction: A mobile application for non-health providers in the Philippines.
8. Erlich, P. (2015). Preparedness of medical personnel for humanitarian missions in the Philippines: A checklist approach.